

Oct.-16/TE/Insem.-40

T.E. (Computer Engineering) (Semester - I)

THEORY OF COMPUTATION

(2012 Pattern)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer either Q1 or Q2, and Q3 or Q4, and Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Assume suitable data if necessary.
- 4) Give suitable examples wherever necessary.

Unit - I

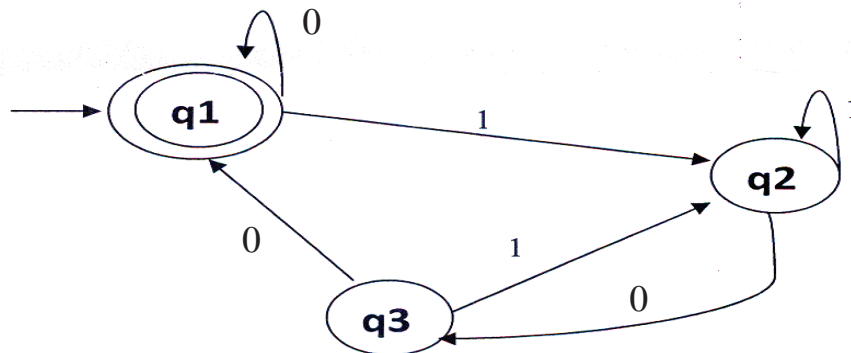
- Q1)** a) Give Regular Expression for the following languages : [6]
- i) $L = \{ x \mid x \in \{ a, b \}^* \text{ and } x \text{ contains exactly two } a\text{'s} \}$
 - ii) $L = \{ a, c, ab, cb, abb, cbb, abbb, cbbb, \dots \}$
 - iii) $L = \{ x \mid x \in \{ a, b \}^* \text{ and } x \text{ is any string that begins in "abb" or "a"} \}$
- b) Design Finite Automata (FA) for accepting strings, over $\Sigma = \{ 0, 1 \}$, with even number of 0's and odd number of 1's. [4]

OR

- Q2)** a) Construct a DFA with reduced states equivalent to the regular expression $10 + (0+11)0^*1$. [6]
- b) Prove the formula [4]
- i) $(r * s^*)^* = (r + s)^*$
 - ii) $(ab)^* \neq a^* b^*$

Unit - II

- Q3)** a) Find the regular expression for the set of strings recognized by the given FA. Use Arden's theorem. [5]



P.T.O.

b) Minimise the DFA given below:

[5]

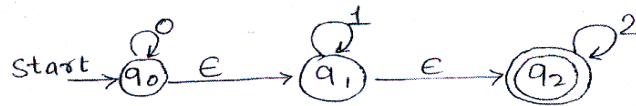
Input Symbols -> States	Next State	
	a	b
-> * 1	3	2
2	4	1
3	5	4
4	4	4
*5	3	2

Initial State: State 1 and Final States: States 1 and 5

OR

Q4) a) Design a Moore machine for computing the 2's complement of a binary number. Convert it into its equivalent Mealy machine. [6]

b) Consider the following NFA with E-transitions. Convert this in to NFA without ϵ - moves. [4]



Unit - III

Q5) a) Give context free grammars for the following languages: [6]

i) $L = \{ x \mid x \in \{ (,) \}^* \text{ with strings having well-formed parentheses (WFP)} \}$

ii) $L = \{ a^m b^n c^{m+n} \mid m, n \geq 0 \}$

b) Explain, with suitable examples, any two applications of context free grammars. [4]

OR

Q6) a) Convert the following CFG into Chomsky Normal Form (CNF): [6]

$S \rightarrow AB$

$A \rightarrow CA \mid \wedge$

$B \rightarrow DB \mid \wedge$

$C \rightarrow 011 \mid 1$

$D \rightarrow 01$

b) Explain graph grammars with a suitable example. [4]

